ESOF 5014: Agile Software Development

Topics

There are four project options available for students to choose from. Each option addresses a real-world problem and is designed to support the application of Agile Software Development practices and Multi-Agent System design. Students who wish to pursue a custom project topic may propose their own idea. A brief written proposal outlining the project scope, objectives, and relevance must be submitted to the Instructor or Teaching Assistant no later than the end of Week 4. Approval is required before proceeding with a custom project.

Groups

All projects must be completed in groups of three (3) students. Each group must consist of either all graduate students or all undergraduate students. Mixed groups are not permitted.

Deliverables

- A **midterm presentation** will take place before Reading Week, where each team will present a progress report.
- A **final presentation** and **final written report** will be required at the end of the term.

Specific dates and presentation schedules will be announced in class and posted on D2L. It is highly recommended that each team meet with the Teaching Assistant on a weekly basis to demonstrate progress and receive feedback.

D2L Portal

Detailed rubrics for all deliverables will be posted on the D2L course portal. Student groups must also be formed through D2L under Other Tools → Groups.

Late Submission Policy

Late presentations are not permitted. In the case of a planned absence, the student must submit a formal request to the Teaching Assistant at least one week in advance. In the case of a medical or emergency situation, the student must notify the Teaching Assistant as soon as they become aware of the issue and provide appropriate documentation. Late submission of the **final project report** will result in a penalty of 15% of the maximum grade **per each day**.

Additional Requirements for Graduate Students

1. Graduate students are expected to demonstrate creativity and innovation in their implementation. This may include:
   - Additional or advanced system functionality
   - Novel approaches to solving the described problems
   - Enhanced modeling, simulation, or evaluation components
2. The final written report must follow the **IEEE two-column conference format** and be prepared using a LaTeX environment.

Option 1

1. Project Title: Smart Home Energy Management System with Multi-Agent Collaboration for Renewable Energy Production, Consumption Prediction, and Peer-to-Peer Energy Trading

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2. Background and Motivation:

The global energy landscape is rapidly transitioning towards renewable energy sources to combat climate change and reduce carbon footprints. In this context, smart homes equipped with renewable energy systems such as solar panels and wind turbines are gaining prominence. However, challenges remain in optimizing energy production, managing consumption, and facilitating efficient peer-to-peer (P2P) energy trading among neighbors.

Multi-agent systems (MAS) offer a promising framework to address these challenges. By deploying autonomous agents with specific roles, the system can dynamically adapt to varying energy demands, optimize resource usage, and enable seamless energy trading. This project aims to design and implement a MAS for smart homes to enhance energy efficiency and foster a collaborative energy ecosystem.

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3. Objectives:

3.1 (20% Mark)

Develop the mathematical model for each agent. In your project, you should have at least five agents as follows:

- Prediction Agent: This agent must be able to predict the energy production and consumption for a house, determine the demand or surplus on an hourly basis.
- Demand Response Agent: This agent is responsible for interacting with the energy grid agent to engage in the demand response and curtailment requests to balance supply demand of energy:
- Behavioral and Segmentation Agent: This agent is responsible for appliance-to-appliance prediction and classification to determine the priority of smart home usage of appliances
- Negotiation Agent: This agent is responsible for energy trading among the neighbors and the power grid. This agent implements state-of-the-art auction theory. Your design should prove all the properties of auction theory for algorithmic game theory design
- Facilitating Agent: This agent serves as the interface and coordination among the agents in the system. You should design the interface of this agent and set the properties of interaction.

3.2 (15% Mark)

Create a peer-to-peer energy trading platform to facilitate secure and efficient energy transactions among neighboring homes.

3.3 (15% Mark)

Evaluate the system's performance in terms of energy efficiency, prediction accuracy, and trading efficacy.

3.4 (20% Mark)

Use Software Agent Development Environment (e.g. JADE) to develop the system. The methodology of designing the project should be Agile.

3.5 (10% Mark)

The students should find datasets on their own. Smart home datasets should include appliance level energy consumption, datasets for solar or wind should include hourly energy production, dataset for power grid should include energy consumption at the lowest granular level.

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4. Deliverables:

4.1 System Architecture:

The proposed system comprises the following components:

- Renewable Energy Sources: Solar panels and/or wind turbines.
- Energy Storage: Batteries for storing surplus energy.
- Multi-Agent System (MAS): Interaction model and protocol
- User Interface: Provides real-time insights and control options for homeowners.

4.2 Mathematical Model for each agent and proofs of auction theories

4.3 Predictive Modeling:

Develop machine learning models, such as LSTM networks or hybrid approaches, to forecast short-term and long-term energy production and consumption based on historical data, weather conditions, and user behavior patterns.

4.4 Peer-to-Peer Trading Platform:

Design a secure (e.g. blockchain-based) platform for secure and transparent energy transactions. You may use smart contracts to automate trading processes based on predefined rules, ensuring fairness and efficiency.

4.5 Simulation and Testing:

Simulate the system in a controlled environment using synthetic and real-world datasets. Evaluate the MAS's performance under various scenarios, including peak demand, renewable energy surplus, and dynamic market conditions.

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5. Expected Outcomes:

1. A functional prototype of a smart home energy management system powered by a multi-agent framework.
2. Accurate predictive models for energy production and consumption.
3. A secure and efficient P2P energy trading platform.
4. Quantitative analysis of the system's benefits, including cost savings, energy efficiency, and reduced reliance on the main grid.

Option 2

1. Project Title: Decentralized Multi-Agent Disaster Relief Coordination System for Resource Allocation, Volunteer Management, and Real-Time Crisis Response

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2. Background and Motivation:

During large-scale disasters (earthquakes, floods, wildfires), centralized emergency response systems often become overwhelmed, slow, or entirely unavailable due to communication failures, infrastructure collapse, or network overload. This significantly delays the delivery of critical resources such as food, water, shelter, and medical aid.

A decentralized multi-agent system (MAS) can increase resilience by allowing autonomous agents—representing volunteers, NGOs, local authorities, drones, shelters, hospitals, and affected communities—to collaborate using distributed negotiation and communication protocols.
By removing single points of failure, the system can self-organize, dynamically adapt to chaos, and continue functioning even when regional connectivity is compromised.

This project aims to design and prototype a decentralized MAS platform that supports resource allocation, volunteer-task matching, real-time situational assessment, and robust coordination during crisis scenarios.

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3. Objectives:

3.1 (20% Mark)

Develop the mathematical model for each agent. Include at least five agents, for example:

- Victim/Request Agent: Represents individuals or communities generating resource requests (food, shelter, medical). Responsible for priority scoring based on severity and urgency.
- Volunteer Agent: Responsible for reporting availability, skills (e.g., medical, logistics), and mobility. Capable of negotiating tasks through distributed protocols.
- Resource Hub Agent: Represents shelters, supply depots, or medical centers. Tracks inventory levels and dispatches resources.
- Negotiation & Routing Agent: Implements distributed task allocation and multi-agent negotiation (e.g., Contract Net Protocol, or auction-based allocation). Includes mathematical proofs for allocation fairness or optimality.
- Coordination/Facilitating Agent: Defines communication interfaces, decentralization protocols, and ensures robust coordination across unstable networks.

3.2 (15% Mark)

Design a decentralized coordination mechanism (e.g., peer-to-peer or blockchain-based) ensuring system continuity even under partial network failure.

3.3 (15% Mark)

Evaluate system performance under simulated disaster scenarios (communication failure, high request load, resource scarcity). Metrics may include response time, fairness, and system resilience.

3.4 (20% Mark)

Implement the MAS using a Software Agent Development Environment (e.g., JADE). Project management must follow Agile methodology (sprints, backlog, user stories, etc.).

3.5 (10% Mark)

Students must find or construct datasets for disaster events, resource needs, volunteer mobility, and supply chain flows. Public datasets (FEMA, UN OCHA, CrisisNLP) may be used.

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4. Deliverables:

4.1 System Architecture: Include key components such as

- Decentralized communication layer (peer-to-peer overlay, blockchain optional)
- Multi-Agent System communication and interaction model
- Data ingestion and situational awareness module
- Volunteer and resource dashboard for monitoring and interaction

4.2 Mathematical Model:

Provide mathematical specifications for each agent, negotiation model, and formal justification of fairness/optimality for the chosen distributed protocols.

4.3 Disaster Simulation Engine:

Develop a simulation environment with multiple crisis scenarios to test the MAS's adaptive behavior.

4.4 Decentralized Coordination Layer:

May include blockchain/smart contracts for immutable logging of resource distribution or volunteer-task assignment.

4.5 Testing & Evaluation:

Perform simulations under different crisis intensities and infrastructures (partial outages, limited volunteers, bottlenecks). Provide quantitative analysis.

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5. Expected Outcomes:

5. A functioning prototype demonstrating decentralized disaster relief coordination.
6. A MAS capable of self-organizing under stress with no single point of failure.
7. Quantitative evidence of improved responsiveness, resilience, and fairness.

Option 3

1. Project Title: Multi-Agent System for Smart City Resource Management and Optimization

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2. Background and Motivation:

Modern smart cities rely heavily on efficient management of interconnected systems such as traffic, public transportation, waste collection, water usage, environmental monitoring, and energy distribution. As urban populations grow, centralized control approaches become increasingly insufficient due to high computational loads, dynamic changes, and the need for real-time decision-making.

Multi-agent systems (MAS) provide a scalable and adaptive solution. By distributing control among autonomous agents—each responsible for specific city functions—the system can better respond to local events, reduce congestion, optimize resource allocation, and coordinate across subsystems. This project aims to design and implement an MAS capable of managing core smart city services while demonstrating intelligent coordination, negotiation, and adaptability.

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3. Objectives

3.1 (20% Mark)

Develop the mathematical model for each agent. Your project should include at least five agents, for example:

- Traffic Signal Agent: Adjusts signal timings based on congestion predictions and prioritization policies.
- Public Transit Agent: Manages bus or tram schedules dynamically according to passenger flow and delays.
- Energy Grid Agent: Allocates electricity across districts, responding to peak load or renewable availability.
- Environmental Monitoring Agent: Detects air quality, noise levels, or pollution and sends alerts to other agents.
- Resource Coordination Agent: Facilitates communication among agents and resolves conflicting goals using negotiation protocols.

3.2 (15% Mark)

Implement inter-agent communication and negotiation mechanisms. Students must design a communication protocol (JADE ACL or custom extension) that allows the agents to:

- Share real-time contextual information
- Negotiate resource allocation
- Resolve conflicts using multi-agent negotiation strategies

3.3 (15% Mark)

Evaluate the system's performance across different city scenarios. Examples include:

- Traffic surges
- Energy shortages or excess
- Air quality deterioration
- Public transit delays
  Students must provide metrics, visualizations, and comparative results.

3.4 (20% Mark)

Develop the system using a Software Agent Development Environment (e.g., JADE) following Agile methodology.

For example:

- Backlog
- Sprints
- Design artifacts
- Burndown charts

3.5 (10% Mark)

Students must find datasets independently.

Possible datasets:

- Open traffic flow datasets
- Public transit logs
- Air quality datasets
- Smart grid datasets
- GIS-based city data

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4. Deliverables

4.1 System Architecture

The system should include:

- MAS with well-defined interaction protocols
- Real-time data ingestion pipeline
- Decision-making modules
- Visual dashboard for city administrators

4.2 Mathematical Models for All Agents

Models should include:

- Prediction models (e.g., time-series, ML-based)
- Optimization formulations
- Multi-agent coordination algorithms

4.3 Coordination and Negotiation Mechanisms

Students should implement:

- Contract-net protocol, auction-based decision-making, or rule-based coordination
- Conflict-resolution algorithms
- Messaging and communication interfaces

4.4 Simulation and Testing

Students must:

- Simulate various city dynamics using real or synthetic datasets
- Analyze system behavior under load, failures, or conflicting goals
- Demonstrate improvements in efficiency, congestion reduction, or resource use

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5. Expected Outcomes

1. A functional multi-agent prototype capable of managing core smart city resources.
2. Intelligent autonomous agents that coordinate and negotiate in real time.
3. Quantitative evaluation demonstrating improvements in city resource efficiency.
4. Potential for extension into smart mobility, IoT integration, or environmental sustainability research.

Option 4

1. Project Title: Multi-Agent Cybersecurity Training Simulator with Gamified Attack and Defense Scenarios

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2. Background and Motivation:

Cybersecurity education often relies on static exercises, theoretical training, or manually designed Capture-the-Flag activities. These approaches fail to capture the dynamic, adversarial, and evolving nature of modern cyber threats.

A multi-agent gamified cybersecurity simulator can provide an interactive learning environment where users defend a network from autonomous attack agents such as phishing bots, brute-force attackers, ransomware spreaders, and insider threat agents.
Players earn points by detecting intrusions, patching vulnerabilities, deploying countermeasures, and collaborating with defensive agents.

This approach offers a hands-on, adaptive, and motivating educational experience with strong potential for research and practical impact.

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3. Objectives:

3.1 (20% Mark)

Develop the mathematical model for at least five agents:

- Phishing Attack Agent: Sends deceptive messages; uses probabilistic targeting.
- Brute-Force Agent: Attempts password cracking with adaptive strategies.
- Malware Propagation Agent: Models infection spread across the network.
- Defender Agent: Patches vulnerabilities, deploys scanners, and assists the user.
- Monitoring/Scoring Agent: Tracks user actions, threat levels, and awards points/achievements.

3.2 (15% Mark)

Design a gamification structure, including:

- XP and leveling
- Timed challenges
- Defense ranking
- Combo rewards for fast remediation
- "Boss-level" attacks (for example ransomware burst, multi-vector attack)

3.3 (15% Mark)

Evaluate the training environment based on:
Detection time, threat mitigation success rate, effectiveness of defensive strategies, and player skill improvement over time.

3.4 (20% Mark)

Build the MAS using JADE or any agent-based library. Project execution must follow Agile principles.

3.5 (10% Mark)

Students must gather or simulate datasets related to network logs, intrusion attempts, vulnerabilities, or malware spread patterns.

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4. Deliverables:

4.1 System Architecture:

Include modules such as:

- Network simulation environment
- Attack and defense agent communication model
- Gamification dashboard
- Threat analytics module

4.2 Mathematical Agent Models:

Define threat propagation formulas, attack success probability, defense utility functions, and scoring metrics.

4.3 Gamification Engine:

Includes missions, achievements, levels, heat-map visualizations, and performance scoring.

4.4 Attack & Defense Scenario Simulation:

Implement multiple scenarios:

- phishing wave
- brute-force burst
- malware spread
- insider attack

4.5 Testing & Results:

Analyze performance of systems and player strategies across multiple trials.

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5. Expected Outcomes:

1. A functional gamified cybersecurity training tool with intelligent attack and defense agents.
2. Demonstration of MAS interactions in adversarial environments.
3. Quantitative evidence that gamification improves learning and detection speed.